

F²ACT-VQA: Fusion-Driven Fact Retrieval with Generative Reranking for Knowledge-Intensive Visual Question Answering

Wooseok Han*
Tsinghua University
Beijing, China
hany21@mails.tsinghua.edu.cn

Wonduk Seo*[†]
University College London
London, United Kingdom
wonduk.seo.26@ucl.ac.uk

Lee Jung-Mok*
KAIST
Daejeon, South Korea
jungmok.lee@kaist.ac.kr

Juhyeon Lee
LG AI Research
Seoul, South Korea
leejuhyeon@lgresearch.ai

Junseo Koh
Peking University
Beijing, China
junseokoh09@gmail.com

Yi Bu[‡]
Peking University
Beijing, China
buyi@pku.edu.cn

Abstract

Knowledge-intensive visual question answering requires identifying the subject in an image and retrieving question-relevant external knowledge. Existing approaches often make a hard entity decision before evidence retrieval or retrieve coarse document-level evidence, making them vulnerable to irreversible entity errors and irrelevant context. In this paper, we propose **F²ACT-VQA**, a novel fusion-driven fact-centric multimodal framework that performs global retrieval over atomic knowledge-base FACTs represented as Subject–Predicate–Object (SPO) triples. Instead of selecting an entity first, **F²ACT-VQA** converts the image–question pair into a structured multimodal retrieval intent and directly matches it against the entire FACT store. Weighted fusion combines the generated query FACT with the image; multimodal reranking and a category-aware inspector then produce a compact, diverse evidence set for grounded answer generation. Experiments on InfoSeek and Encyclopedic VQA demonstrate the effectiveness of **F²ACT-VQA** in both fact retrieval and downstream question answering.¹

CCS Concepts

• **Information systems** → **Query representation**; **Document structure**; **Question answering**; **Multimedia and multimodal retrieval**.

Keywords

visual question answering, multimodal retrieval, retrieval-augmented generation, fact-centric retrieval, generative reranking

1 Introduction

Visual Question Answering (VQA) aims to answer natural-language questions about visual inputs by connecting a user’s intent with relevant visual evidence [2, 3, 9, 14]. This capability supports applications such as visual assistance, information access, and human–AI interaction [13, 16, 47]. The emergence of Multimodal Large Language Models (MLLMs) has substantially advanced VQA through large-scale vision–language pretraining and instruction tuning [2, 25–27, 31]. However, many real-world questions require knowledge

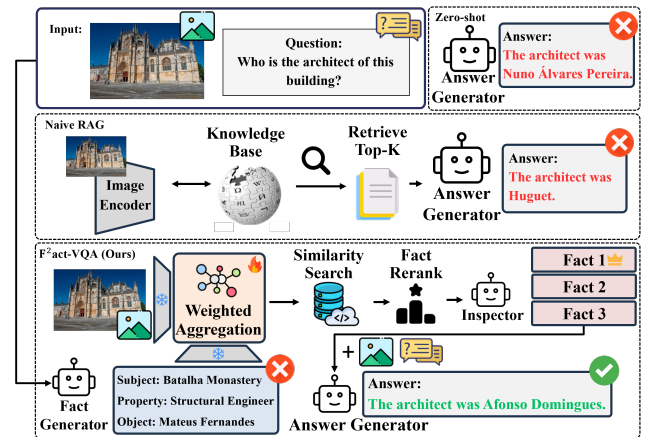


Figure 1: Overview of **F²ACT-VQA**, which retrieves and reranks atomic FACTs using a structured multimodal query for evidence-grounded answer generation.

that is not directly observable in an image. When a question concerns a long-tailed entity or knowledge that is absent or outdated in model parameters, MLLMs may fail to answer correctly or generate plausible but factually unsupported responses [18, 35, 37]. Their reliance on internal parametric knowledge therefore limits their reliability in knowledge-intensive VQA.

Following foundational work on text-based Retrieval-Augmented Generation (RAG) [24], recent work has incorporated external retrieval to ground answer generation [7, 29, 30, 56]. These systems retrieve evidence from knowledge bases or document collections using the query image, the textual question, image-derived descriptions, or joint multimodal representations, and provide the retrieved context to an MLLM together with the original input [7, 30, 39, 42, 56]. More recent approaches further refine candidate evidence through multimodal reranking, explicit entity identification, or reflective relevance assessment [12, 17, 34, 57].

Despite this progress, three limitations remain. (1) Existing retrieval formulations often fail to preserve both visual entity uncertainty and question-specific factual intent. Approaches that compress the image and question into a single representation can conflate these complementary signals, while entity-first pipelines make

*These authors contributed equally to this work.

[†]Incoming PhD student.

[‡]Corresponding Author.

¹Our code is available at: <https://github.com/hws2002/f2act-vqa-official>.

an early discrete commitment that irrevocably constrains subsequent retrieval. As a result, both designs may hinder direct matching between a multimodal query and fine-grained entity–relation evidence. (2) Retrieval is typically performed over coarse evidence units (e.g., documents or sections). Even when the correct entity is retrieved, these units may omit the specific predicate and answer required by the question and may introduce substantial irrelevant context. (3) Finally, selecting only the top-ranked candidates can yield redundant evidence, allowing near-duplicate facts to crowd out complementary information needed for answer generation.

To address these limitations, we propose **F²ACT-VQA**, a fusion-driven, fact-centric multimodal framework that directly retrieves atomic knowledge-base facts represented as Subject–Predicate–Object (SPO) triples. Building on prior triplet-based representations for knowledge-base construction and question answering [1, 23, 53], **F²ACT-VQA** augments them with multimodal fusion, reranking, and diversity-aware evidence selection. As illustrated in Figure 1, it first reformulates an image I and question Q into a query $\text{FACT } \hat{f} = (\hat{s}, \hat{p}, \hat{o})$ that captures the depicted entity, requested relation, and expected answer semantics. We then fuse the visual embedding of I with the textual embedding of $\hat{f}$ to query a fact store in which each entry combines an atomic FACT with the corresponding subject’s Wikipedia lead section. Dense retrieval returns the top- K candidate FACTs, which are subsequently reranked by a multimodal MLLM. We fuse the retriever and reranker rankings via Reciprocal Rank Fusion (RRF) to leverage their complementary relevance signals. Finally, we apply a category-aware inspector that groups candidates into P semantic categories and retains the highest-ranked FACT per group. Conditioned on I , Q , and the resulting top- P facts, an MLLM generates the final answer.

We evaluate **F²ACT-VQA** on two representative knowledge-intensive VQA benchmarks, InfoSeek [10] and Encyclopedic VQA (E-VQA) [37]. Across retrieval metrics and downstream answer accuracy, our approach consistently outperforms competitive baselines, demonstrating the benefit of direct fact retrieval. Our main contributions are summarized as follows:

- We introduce a structured multimodal retrieval formulation that performs global search over the entire knowledge base for atomic, answer-bearing SPO FACTs, avoiding the irreversible bottleneck induced by hard entity selection.
- We develop weighted fusion over visual features and structured FACT representations (with Wikipedia-enriched KB context), and refine retrieved candidates via multimodal reranking and Reciprocal Rank Fusion (RRF).
- We propose a category-aware evidence selection strategy to reduce redundancy, and evaluate the resulting framework on InfoSeek and E-VQA at both the retrieval level and the downstream question-answering level.

2 Related Work

2.1 MLLMs for Knowledge-Intensive VQA

Recent state-of-the-art MLLMs have shown remarkable performance across diverse tasks, including visual question answering (VQA) [2, 3, 9, 14, 22, 25, 27, 60]. Despite this progress, MLLMs

often struggle with hallucinations due to their reliance on parametric memory [35, 50, 54]. To mitigate this issue, a range of methods [4, 21, 24, 40] adopt retrieval-augmented approaches that integrate relevant external knowledge into MLLMs. Several benchmarks [10, 20, 36, 37, 44, 53] have been introduced to evaluate VQA systems that require external knowledge. OK-VQA, A-OKVQA, and FVQA [36, 44, 53] contain diverse questions that require additional knowledge beyond the image and question, but still grounded in common-sense knowledge that strong MLLMs can often answer directly. InfoSeek [10] and E-VQA [37] are constructed from Wikipedia and require finer-grained encyclopedic knowledge. Accordingly, most methods retrieve relevant Wikipedia *sections* as evidence. While many works use sections or passages as the basic retrieval unit [10, 12, 37, 56, 57], other work explores structured knowledge triplets for knowledge-base storage and retrieval-augmented generation, enabling finer-grained evidence retrieval [1, 23, 53]. Building on this line of work, we propose **F²ACT-VQA**, a knowledge-intensive VQA framework that augments Subject–Predicate–Object (SPO) triplets with multimodal fusion, fact reranking, and diversity-aware evidence selection for fact-grounded answer generation.

2.2 Multimodal Knowledge Retrieval for KB-VQA

Knowledge-based Visual Question Answering (KB-VQA) requires Vision-Language Models (VLMs) to incorporate external knowledge to answer questions, motivating various approaches for knowledge retrieval and integration [12, 15, 20, 33, 56, 57]. One line of work converts visual information into textual descriptions, such as image captions, and uses them to retrieve relevant knowledge [29, 41]. More recent approaches directly leverage multimodal information for retrieval. EchoSight [56] and OMGM [57] adopt a coarse-to-fine pipeline consisting of entity retrieval followed by section reranking. EchoSight first identifies relevant entities through visual-only search and then reranks their sections using the image and question. Similarly, OMGM retrieves entities by matching the query image with entity summaries using a pretrained VLM, followed by multimodal reranking of the retrieved sections. Our method, **F²ACT-VQA**, performs multimodal retrieval by explicitly decomposing each knowledge fact into subject, predicate, and object (SPO) components. The similarity scores computed for each component are then fused to obtain the final relevance score.

3 Methodology

Existing knowledge-based VQA pipelines often commit to an entity before evidence retrieval, causing early identification errors to propagate downstream. We propose **F²ACT-VQA**, a training-free multimodal RAG framework that represents multimodal queries and knowledge-base facts in a shared FACT space, enabling global retrieval of atomic answer-bearing evidence without hard entity selection. An overview of **F²ACT-VQA** is provided in Figure 2.

3.1 Overview

Given an image I and question Q , our objective is to generate an answer $\hat{A}$ grounded in an external knowledge base, which we

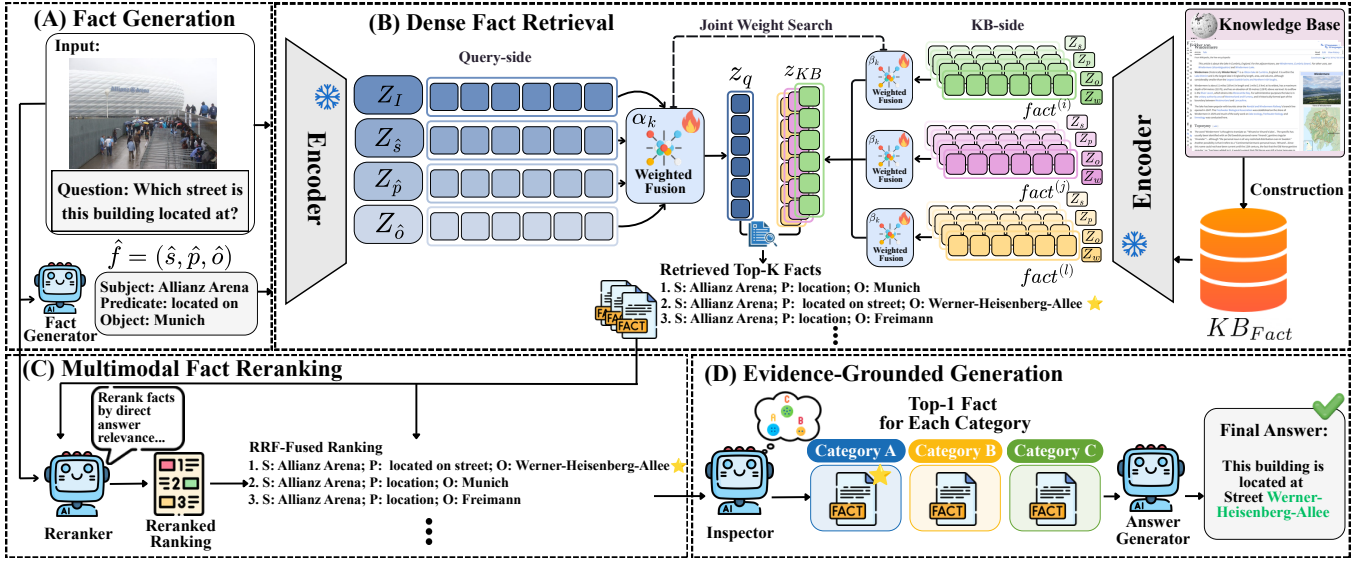


Figure 2: Overview of F²ACT-VQA. The framework comprises four stages: (a) FACT Generation reformulates the image-question pair into a query-side Subject-Predicate-Object (SPO) FACT; (b) Dense FACT Retrieval performs weighted multimodal matching between the image-enriched query FACT and Wikipedia-enriched knowledge-base FACTs to retrieve the top- K candidates; (c) MLLM Reranking reorders candidates by relevance to the image and question and fuses this ranking with the initial retrieval ranking via Reciprocal Rank Fusion (RRF); and (d) Fact-Grounded Answering groups the ranked FACTs into semantic categories, selects the highest-ranked FACT per category, and conditions the answer generator on the resulting diverse evidence set.

represent as a fact store $\mathcal{F} = \{f_i\}_{i=1}^N$ containing N atomic FACTs. Each FACT is expressed in Subject-Predicate-Object (SPO) form:

$$f = (s, p, o), \quad (1)$$

where s , p , and o denote the subject, predicate, and object of f , respectively. This representation provides a compact retrieval unit that distinguishes the entity of interest, the requested relation, and its associated value.

Operating in this shared FACT space, F²ACT-VQA constructs a query FACT from (I, Q) and encodes both the multimodal query and KB-side FACTs through component-wise embedding fusion. Global dense retrieval identifies the top- K candidate FACTs, which are refined by a multimodal MLLM reranker. The category-aware Inspector G_{ins} then selects a diverse set of P FACTs, from which the answer generator G_{ans} produces the final grounded answer $\hat{A}$.

3.2 Dense Fact Retrieval

Our dense retrieval operates directly in the shared FACT space, aiming to retrieve fine-grained, answer-bearing evidence without relying on an intermediate entity retrieval step. Motivated by prior work showing that appropriately weighted fusion of representations can improve retrieval effectiveness in textual and multimodal settings [32, 45, 55, 58], we encode the multimodal query and KB-side FACTs component-wise and combine their signals using side-specific fusion weights. The resulting representations occupy the same embedding space, enabling direct relevance matching between the query and every FACT in the knowledge base.

Query fact generation. To align the multimodal query with the shared FACT space, we construct a query FACT in Subject-Predicate-Object (SPO) form from the image I and question Q , thereby obtaining a compact, structured representation for fact-level retrieval. Specifically, we use a frozen MLLM, denoted by G_{fact} , to generate the natural-language query FACT. Recent studies have shown that LLMs can reliably extract structured relational triplets in zero- and few-shot settings [38, 51], supporting their use for query FACT generation. We use the following prompt:

Prompt for Query Fact Generation

Given an image of a real-world entity and a question about it, generate a concise Subject-Predicate-Object FACT that captures the information needed to answer the question. Specifically...

A generated fact is expressed as

$$\hat{f} = (\hat{s}, \hat{p}, \hat{o}) = G_{\text{fact}}([I, Q])$$

Query fact encoding. To construct the query representation, the image and each textual component of the generated fact $\hat{f}$ are encoded separately using frozen CLIP encoders. Let E_v and E_t denote the visual and text encoders of CLIP, respectively. The resulting component-level embeddings are given by

$$\mathbf{z}_I = E_v(I), \quad \mathbf{z}_{\hat{s}} = E_t(\hat{s}), \quad \mathbf{z}_{\hat{p}} = E_t(\hat{p}), \quad \mathbf{z}_{\hat{o}} = E_t(\hat{o}) \quad (2)$$

The resulting query-side embeddings are combined using fusion weights α_k to form a unified query embedding:

$$\mathbf{z}_q = \sum_{k \in \{I, \hat{s}, \hat{p}, \hat{o}\}} \alpha_k \mathbf{z}_k, \quad (3)$$

where $\mathbf{z}_q$ denotes the query embedding, and α_k controls the contribution of the k -th query component.

KB fact encoding. Each knowledge-base entry is primarily represented by its atomic FACT components: subject s , predicate p , and object o . These components enable fine-grained matching with the query-side FACT. We additionally use the Wikipedia lead section w of the subject’s article, which provides auxiliary context for entity disambiguation. All textual views are encoded using the same frozen CLIP text encoder E_t :

$$\mathbf{z}_w = E_t(w), \quad \mathbf{z}_s = E_t(s), \quad \mathbf{z}_p = E_t(p), \quad \mathbf{z}_o = E_t(o). \quad (4)$$

Analogously, the resulting KB-side embeddings are combined using fusion weights β_k :

$$\mathbf{z}_{KB} = \sum_{k \in \{w, s, p, o\}} \beta_k \mathbf{z}_k. \quad (5)$$

where $\mathbf{z}_{KB}$ denotes the fused representation of a KB FACT, and β_k controls the contribution of each KB-side component. We select the query- and document-side fusion weights α and β via grid search on the InfoSeek training split.

Dense Search. The fused query and KB representations establish a common embedding space for global fact-level retrieval. Each KB FACT is directly scored against the multimodal query over the entire FACT store, enabling relevance to be assessed at the level of fine-grained evidence. We specifically compute the cosine similarity between the fused query and KB representations:

$$S_{\text{ret}}(\mathbf{z}_q, \mathbf{z}_{KB}) = \text{sim}(\mathbf{z}_q, \mathbf{z}_{KB}) = \frac{\mathbf{z}_q^\top \mathbf{z}_{KB}}{\|\mathbf{z}_q\|_2 \|\mathbf{z}_{KB}\|_2}. \quad (6)$$

Accordingly, all KB FACTs are ranked by S_{ret} , and the top- K candidates are retained as the initial retrieval set:

$$\mathcal{F}_{\text{ret}} = \left\{ f_{\text{ret}}^{(1)}, f_{\text{ret}}^{(2)}, \dots, f_{\text{ret}}^{(K)} \right\}. \quad (7)$$

The resulting candidate pool can retain evidence associated with multiple plausible subjects and relations, leaving residual visual and semantic ambiguity to be resolved by the multimodal reranker.

3.3 Multimodal Fact Reranking

Dense retrieval narrows the search space to potentially relevant FACTs, yet ambiguity may remain among visually similar entities or semantically related facts. To further refine $\mathcal{F}_{\text{ret}}$, a multimodal MLLM reranker R_θ evaluates each candidate $f_{\text{ret}}^{(i)} \in \mathcal{F}_{\text{ret}}$. Given the original image–question pair $[I, Q]$, the reranker evaluates each candidate $f_{\text{ret}}^{(i)}$ according to whether it directly provides the knowledge required to answer the question. The following prompt is used for this assessment:

Prompt for Multimodal Fact Reranking

... Retrieve FACTs that directly provide the knowledge needed to answer the question, considering both subject–entity alignment and the relevance of the predicate–object information to the question.

For each candidate, the MLLM reranker produces logits for the YES and NO relevance labels: $\left[\ell_{\text{Yes}}^{(i)}, \ell_{\text{No}}^{(i)} \right] = R_\theta([I, Q], f_{\text{ret}}^{(i)})$

A softmax over the two logits yields the multimodal reranking score S_{rerank} , defined as the probability assigned to the YES label:

$$S_{\text{rerank}}([I, Q], f_{\text{ret}}^{(i)}) = \text{softmax}\left(\left[\ell_{\text{Yes}}^{(i)}, \ell_{\text{No}}^{(i)} \right]\right)_{\text{Yes}}. \quad (8)$$

Score Aggregation. To combine the complementary relevance signals from dense retrieval and multimodal reranking without calibrating their different score scales, we adopt reciprocal-rank fusion (RRF):

$$S_{\text{RRF}}^{(i)} = \frac{1}{k + r_{\text{ret}}^{(i)}} + \frac{1}{k + r_{\text{rerank}}^{(i)}}. \quad (9)$$

Here, $r_{\text{ret}}^{(i)}$ and $r_{\text{rerank}}^{(i)}$ denote the descending ranks induced by $S_{\text{ret}}^{(i)}$ and $S_{\text{rerank}}^{(i)}$, respectively, within the same top- K candidate pool, while k denotes the RRF rank-offset constant.

3.4 Evidence-Grounded Generation

Category-Aware Inspector. The fused RRF scores define an ordered candidate sequence

$$\mathcal{F}_{\text{RRF}} = \left(f_{\text{RRF}}^{(1)}, f_{\text{RRF}}^{(2)}, \dots, f_{\text{RRF}}^{(K)} \right), \quad (10)$$

where candidates are ordered according to their RRF scores. RRF provides a relevance-oriented ranking but does not explicitly model diversity among the retrieved evidence. To obtain broader evidence coverage while retaining the RRF ordering as a relevance prior, we introduce a category-aware Inspector G_{ins} .

Given the original image I , query Q , and $\mathcal{F}_{\text{RRF}}$, the Inspector constructs P evidence categories and selects one representative FACT from each category, while taking the original RRF ordering into account.²

Prompt for Category-Aware Inspector

... Using the image, question, and K ranked facts, create P broad, mutually exclusive evidence categories. Assign each fact to one category and select its best representative, considering the original relevance order.

Formally, the Inspector returns a category name c_p and a representative FACT index j_p for each category:

$$\{(c_p, j_p)\}_{p=1}^P = G_{\text{ins}}(I, Q, \mathcal{F}_{\text{RRF}}), \quad (11)$$

where all category names and selected indices are distinct. The selected indices define the final evidence set

²We study the effect of the category-aware Inspector in Section 4.3.3.

$$\mathcal{F}_{\text{ins}} = \left(f_{\text{RRF}}^{(j_1)}, f_{\text{RRF}}^{(j_2)}, \dots, f_{\text{RRF}}^{(j_p)} \right). \quad (12)$$

Answer Generation. Conditioned on the original image I , question Q , and the selected evidence $\mathcal{F}_{\text{ins}}$, the answer generator G_{ans} produces the final answer:

$$\hat{A} = G_{\text{ans}}([I, Q], \mathcal{F}_{\text{ins}}). \quad (13)$$

4 Experiments

4.1 Setup

4.1.1 Datasets. We evaluate **F²ACT-VQA** on two representative knowledge-intensive VQA benchmarks, InfoSeek [10] and Encyclopedic VQA (E-VQA) [37].

InfoSeek contains 1.35M image-question-answer triplets covering 11K Wikipedia entities. Following the evaluation setup used in prior work [17, 34, 56, 57], we use the same 71,335 question pairs and a 100K-entity Wikipedia knowledge base. Following prior FACT-construction approaches [43, 48], **F²ACT-VQA** builds its FACT knowledge base over the same 100K-entity universe, yielding approximately 1.3M FACTs.

To further assess the effectiveness of **F²ACT-VQA** beyond InfoSeek, we evaluate on E-VQA, which contains more than 1M triplets derived from 221K question-answer pairs across 16.7K fine-grained subject categories. From the official test split, we randomly construct five mutually disjoint subsets of 500 instances each, yielding 2,500 unique evaluation instances in total while preserving the original proportions of single-hop and two-hop questions. We augment the same 100K-entity Wikipedia pool with 1,850 additional entities appearing in the official test split. Since E-VQA does not provide corresponding SPO annotations, we derive FACTs for these additional entities from the benchmark-provided evidence sections following established knowledge-triple extraction approaches [19, 43, 59] and represent them under the same Subject-Predicate-Object (SPO) schema used for InfoSeek.

4.1.2 Models Used.

Retrieval and reranking models. For a controlled comparison at the first-stage retrieval step, all retrieval-augmented baselines and **F²ACT-VQA** use the same frozen EVA-CLIP image and text encoders [49]. Among baselines, Naive-RAG-BGE uses bge-reranker-v2-m3 [8]; EchoSight uses a trained BLIP-2 Q-Former (also reused by CC-VQA); IBA performs MLLM-based entity identification followed by BGE section reranking; and OMGM uses its trained Q-Former followed by BGE section filtering.

To match baseline reranking strategies, we evaluate **F²ACT-VQA** with both an encoder-based and a decoder-based reranker, using bge-reranker-v2-m3 [8] and Qwen3-VL-Reranker-8B [28], respectively. We additionally include Qwen3-VL-Reranker-2B in the reranker ablation to examine scale effects.³

Answerer. We evaluate all methods with three answerer backbones, comprising two 8B open-source models, Qwen3-VL-8B [5] and LLaVA-MORE-8B [11], and one API-based model, GPT-5-mini [46]. The decoding temperature is set to 0 for all models.

³Detailed reranker comparisons are presented in Section 4.3.2.

4.1.3 Baselines. We compare **F²ACT-VQA** with three categories of baselines, including: (i) Zero-shot LLMs, (ii) Naive RAG baselines, and (iii) Knowledge-Intensive VQA baselines.

(i) *Zero-shot LLMs (without external retrieval).* Both **Zero-shot LLM** and **Zero-shot MLLM** answer without external evidence, and only the MLLM additionally takes the query image as input.

(ii) *Naive RAG baselines.* **Naive-RAG-LEAD** and **Naive-RAG-BGE** retrieve candidate entities using EVA-CLIP. **Naive-RAG-LEAD** answers directly without reranking, whereas **Naive-RAG-BGE** reranks the retrieved evidence before generating an answer.

(iii) *Knowledge-Intensive VQA baselines.* **EchoSight** [56] retrieves candidate entities by image similarity and reranks their Wikipedia sections with a trained BLIP-2 Q-Former. **OMGM** [57] adopts a coarse-to-fine pipeline combining image-to-text entity retrieval, trained multimodal reranking, and textual section reranking. **IBA** [34] uses an MLLM to identify the depicted entity from retrieved candidates and then reranks its sections with a text reranker. **CC-VQA** [17] reranks retrieved Wikipedia sections with a multimodal reranker (as in EchoSight) and applies Visual-Centric Contextual Conflict Reasoning (VCCR) to construct conflict-aware context for answer generation.

4.1.4 Implementation Details. All **F²ACT-VQA** results use a fixed configuration. We select the query- and KB-side fusion weights via grid search on the InfoSeek training split, varying each weight from 0.05 to 0.85 in 0.05 increments, and use the setting with the highest average Entity Recall@1 across the three FACT generators.⁴

For a controlled comparison, all methods retrieve 20 candidates and provide 3 evidence units to the final answerer. Baselines retrieve entities and use Wikipedia sections as evidence, whereas **F²ACT-VQA** retrieves and uses atomic FACTs.

For EchoSight, OMGM, IBA, and CC-VQA, we follow the public implementations without additional fine-tuning and use the same three MLLMs for answer generation. CC-VQA retains only Visual-Centric Contextual Conflict Reasoning, while OMGM omits generator training.

4.1.5 Evaluation Metrics. We evaluate **F²ACT-VQA** at both retrieval and answer-generation levels. For retrieval, we report Entity Recall@ K , measuring whether the ground-truth entity appears in the top- K candidates. For answer generation, we follow official protocols: InfoSeek uses exact-match accuracy for string/temporal questions and relaxed accuracy for numerical questions on the unseen-question (UQ) and unseen-entity (UE) splits, plus the official aggregate (harmonic-mean) score; E-VQA uses BERT Matching (BEM) accuracy [6], reported for single-hop (SH) and two-hop (TH) questions.

4.1.6 Hardware Specification. We run Qwen3-VL and LLaVA-MORE on two NVIDIA A100 GPUs, each with 40 GB of memory, and run GPT-5-mini through the OpenAI API.

4.2 Main Experiment Results

Table 1 reports InfoSeek results across three answerer backbones. Zero-shot LLMs underperform because they cannot condition on the image, while zero-shot MLLMs benefit from visual grounding

⁴Grid-search results and the selected weights are provided in Section 4.3.1.

Table 1: Answer accuracy (%) on InfoSeek across three answerer backbones. All retrieval-based rows share EVA-CLIP-8B as the first-stage retriever; later reranking modules are method-specific. Blue rows denote our method: *Ours* (BGE Reranker) uses bge-reranker-v2-m3 [8], and *Ours* (Qwen Reranker) uses Qwen3-VL-Reranker-8B [28]. Statistical significance is assessed using paired *t*-tests with Holm–Bonferroni correction; * denotes a significant improvement over the strongest baseline in the corresponding column ($p < 0.05$). Ret. FT denotes that the retriever is fine-tuned, and Rerank. FT denotes that the reranker is fine-tuned. – denotes N/A; the best result is “bold,” and the second-best is “underlined.” Off. denotes the official split.

Approach	Ret. FT	Rerank. FT	LLaVA-MORE-8B			Qwen3-VL-8B			GPT-5-mini		
			Unseen Query	Unseen Entity	Off.	Unseen Query	Unseen Entity	Off.	Unseen Query	Unseen Entity	Off.
Zero-shot LLMs											
Zero-shot LLM	–	–	2.41	2.30	2.35	4.72	3.24	3.84	2.11	1.96	2.03
Zero-shot MLLM	–	–	9.25	8.43	8.82	20.40	18.35	19.32	19.06	18.62	18.84
Naive RAG Baselines											
Naive-RAG-Lead	×	–	24.60	25.25	24.92	27.35	27.67	27.51	32.14	31.50	31.82
Naive-RAG-BGE	×	×	21.66	20.20	20.90	25.19	25.25	25.22	25.03	24.48	24.75
Knowledge-Intensive VQA											
EchoSight-rerank (2024)	×	✓	33.19	31.70	32.43	33.73	34.12	33.92	35.87	35.51	35.69
OMGM (2025)	×	✓	38.28	37.20	37.73	38.66	37.56	38.10	39.41	39.60	39.50
CC-VQA (2026)	×	✓	31.93	29.94	30.90	33.89	32.43	33.14	31.17	29.59	30.36
IBA (2026)	×	×	35.20	33.90	34.54	34.84	36.69	35.74	37.05	37.05	37.05
Ours (BGE Reranker)	×	×	53.21*	50.01*	51.56*	57.76*	56.95*	57.35*	61.86*	62.38*	62.12*
Ours (Qwen Reranker)	×	×	58.17*	56.13*	57.13*	60.30*	60.06*	60.18*	62.86*	63.81*	63.33*

but remain limited by parametric knowledge. Naive RAG improves accuracy by incorporating retrieved Wikipedia evidence; however, retrieval driven largely by visual similarity can recover the entity but miss the question-specific relation. EchoSight [56] and OMGM [57] partially address this issue via reranking and coarse-to-fine refinement, but still rely on entity- or document-level evidence. IBA [34] commits to an explicit entity prediction before evidence retrieval, making the pipeline sensitive to early errors. CC-VQA [17] adds conflict-aware reasoning, but its effectiveness depends on the relevance and granularity of the retrieved context.

F²ACT-VQA achieves the best performance across all InfoSeek configurations. With the Qwen reranker, **F²ACT-VQA** improves the official score over OMGM by 19.40, 22.08, and 23.83 percentage points for LLaVA-MORE-8B, Qwen3-VL-8B, and GPT-5-mini, respectively, with consistent gains on both unseen-query and unseen-entity splits. Notably, the BGE-reranker variant also yields significant improvements, indicating that the gains are not solely due to MLLM reranking and that global retrieval of atomic FACTs is effective across reranker choices. Switching from BGE to the Qwen reranker provides additional gains of 5.57, 2.83, and 1.21 points, underscoring the benefit of directly modeling image–question–FACT relevance. Since all retrieval-based methods share EVA-CLIP-8B for first-stage retrieval, these improvements reflect our fact-level retrieval and refinement rather than a stronger encoder.

Generalization to E-VQA. We further assess whether **F²ACT-VQA** generalizes beyond the structured FACTs and fusion configuration established on InfoSeek. For E-VQA, which lacks an equivalent atomic fact inventory, we derive and verify SPO-form FACTs from Wikipedia documents using the same representation as InfoSeek. Notably, we keep the InfoSeek-selected fusion weights fixed, and the resulting gains on E-VQA suggest that our weighted aggregation generalizes across benchmarks. Table 2 reports the resulting answer accuracy: **F²ACT-VQA** performs best on both single-hop and two-hop questions. Depending on the answerer

Table 2: Answer accuracy (%) on E-VQA. We use the same experimental settings as Table 1. SH and TH denote single-hop and two-hop questions, respectively.

Approach	LLaVA-MORE-8B		Qwen3-VL-8B		GPT-5-mini	
	SH	TH	SH	TH	SH	TH
Zero-shot LLMs						
Zero-shot LLM	13.82	14.15	16.63	16.24	16.67	15.78
Zero-shot MLLM	13.97	10.44	20.20	19.03	26.92	20.19
Naive RAG Baselines						
Naive-RAG-BGE	26.58	17.63	30.98	19.03	29.77	22.97
Naive-RAG-Lead	22.81	19.95	25.81	19.03	27.65	23.67
Knowledge-Intensive VQA						
EchoSight-rerank (2024)	38.28	17.40	45.14	17.87	44.61	23.20
OMGM (2025)	39.97	19.26	46.83	19.95	47.70	25.52
IBA (2026)	39.15	18.33	46.01	19.03	45.43	24.36
CC-VQA (2026)	37.46	16.47	43.16	16.94	43.79	21.58
Ours (BGE Reranker)	52.01*	25.29*	58.87*	26.45*	59.74*	32.02*
Ours (Qwen Reranker)	53.07*	26.91*	59.45*	28.31*	60.90*	33.18*

backbone, the Qwen-reranker variant improves over the strongest baseline by 13.2 percentage points (single-hop) and 8.36 points (two-hop). These gains indicate that both the FACT representation and the InfoSeek-selected fusion weights transfer effectively across benchmarks.

Entity Retrieval Performance. Figure 3 shows the entity retrieval performance of **F²ACT-VQA** with other baselines. On InfoSeek, **F²ACT-VQA** achieves higher Entity Recall@1 and Recall@3, with the best configuration reaching 66.5% and 78.3%, respectively. Similar improvements are observed on E-VQA, where **F²ACT-VQA** reaches 52.00% Recall@1 and 54.78% Recall@3. These results show that fact-level retrieval with multimodal reranking better preserves ground-truth entities among top-ranked evidence than entity- and section-level retrieval.⁵

⁵Additional analyses of initial entity recall before method-specific reranking, downstream answer accuracy on matched Top-1 hit/miss subsets, and entity coverage under matched evidence budgets will be provided in a forthcoming technical supplement.

Table 3: Entity Recall@1 (%) for fusion-weight configurations on InfoSeek. (w/o) Image excludes the query image; component-only settings retain the image with only the corresponding FACT component; Equal-weighting uses uniform weights, and Grid-selected denotes the best grid-searched configuration.

Weight setting	Query ($\alpha_I, \alpha_s, \alpha_p, \alpha_o$)	Document ($\beta_w, \beta_s, \beta_p, \beta_o$)	Entity R@1
(w/o) Image	(0, .33, .33, .33)	(.25, .25, .25, .25)	26.63
Subject-only	(.50, .50, 0, 0)	(0, 1, 0, 0)	30.36
Predicate-only	(.50, 0, .50, 0)	(0, 0, 1, 0)	29.33
Object-only	(.50, 0, 0, .50)	(0, 0, 0, 1)	18.38
Equal-weighting	(.25, .25, .25, .25)	(.25, .25, .25, .25)	32.24
Grid-selected	(.65, .10, .20, .05)	(.30, .10, .55, .05)	58.26

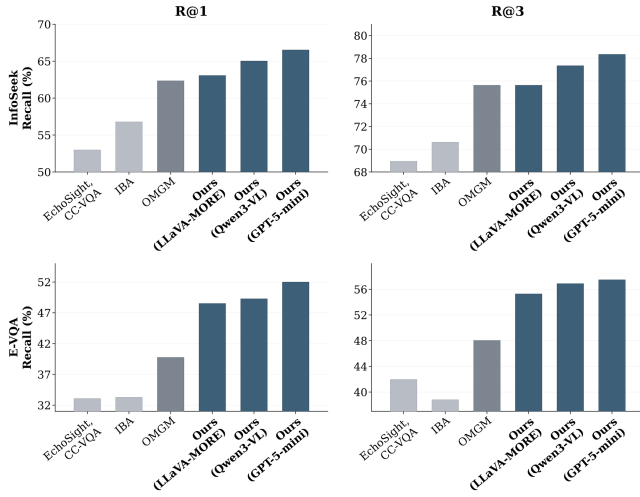


Figure 3: Entity retrieval performance on InfoSeek and E-VQA. We report Entity Recall@1 and Recall@3 for knowledge-intensive VQA baselines and our method across different query-side SPO generators. Baseline results are measured after section reranking and ours are measured after multimodal fact reranking.

4.3 Ablation Studies

4.3.1 Analysis of Retrieval Fusion Weights. Table 3 analyzes the effect of component weighting on dense FACT retrieval. Removing the image or relying on individual SPO components underperforms equal weighting, whereas the grid-selected configuration improves Entity Recall@1 from 32.24% to 58.26% and assigns larger weights to the image (0.65) and KB-side predicate (0.55). Figure 4 further confirms this pattern, with high-recall regions concentrated around larger Z_I and Z_p weights. Although assigned a small weight, the object provides complementary value-level information for disambiguating otherwise similar FACTs.

4.3.2 Effectiveness of Fact-Aware MLLM Reranking. To evaluate the effectiveness of MLLM-based reranking, we conduct an ablation study over different rerankers and model sizes, comparing BGE [8] with Qwen3-VL-Reranker (2B/8B) [28]. As shown in Table 4, BGE

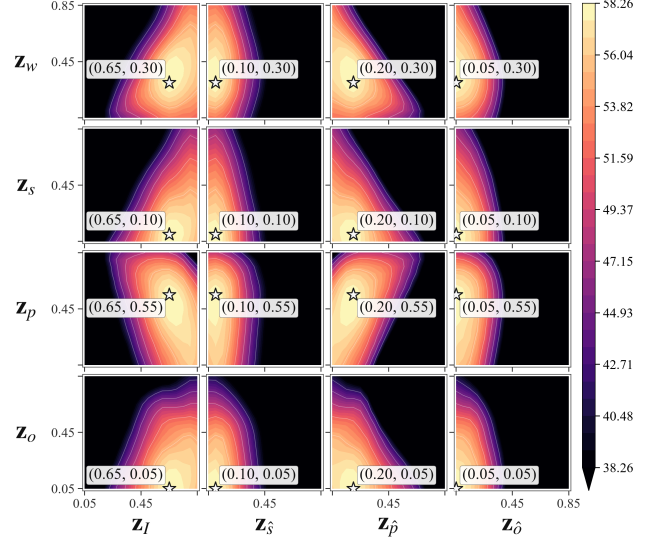


Figure 4: Pairwise embedding-weight grid search on the InfoSeek training set. Each panel shows the best entity Recall@1 for a query- and KB-side weight pair, maximizing over weights. Stars mark the optimum in each panel.

Table 4: Reranker ablation on InfoSeek and E-VQA using Entity Recall. Within each SPO-generator block, the best result is bolded and the second-best is underlined. Q3-VL-R2B and Q3-VL-R8B denote Qwen3-VL-Reranker-2B and Qwen3-VL-Reranker-8B, respectively.

Reranker	R@1		R@3		R@5		R@10	
	InfoSeek	E-VQA	InfoSeek	E-VQA	InfoSeek	E-VQA	InfoSeek	E-VQA
LLaVA-MORE-generated SPO								
BGE	51.42	42.56	67.47	52.04	74.89	56.16	83.00	61.68
Q3-VL-R2B	<u>58.47</u>	<u>43.80</u>	<u>72.48</u>	<u>53.28</u>	<u>78.04</u>	<u>57.32</u>	<u>83.93</u>	<u>62.48</u>
Q3-VL-R8B	63.06	48.52	75.63	55.28	79.92	58.52	84.49	62.72
Qwen3-VL-8B-generated SPO								
BGE	54.51	44.48	70.53	53.80	77.35	57.92	84.59	62.32
Q3-VL-R2B	<u>60.35</u>	<u>45.84</u>	<u>74.18</u>	<u>55.44</u>	<u>79.45</u>	<u>59.00</u>	<u>85.13</u>	<u>62.68</u>
Q3-VL-R8B	65.02	49.28	77.35	56.88	81.38	59.80	85.67	62.88
GPT-5-mini-generated SPO								
BGE	57.10	45.52	72.24	55.44	78.47	58.88	85.26	62.48
Q3-VL-R2B	<u>62.40</u>	<u>47.08</u>	<u>75.65</u>	<u>56.16</u>	<u>80.58</u>	<u>59.44</u>	<u>86.03</u>	<u>62.76</u>
Q3-VL-R8B	66.52	52.00	78.35	57.48	82.13	59.96	86.43	63.00

consistently underperforms the multimodal rerankers on both InfoSeek and E-VQA, demonstrating the benefit of incorporating visual grounding. Moreover, the 8B variant consistently outperforms the 2B variant, suggesting that increased model capacity further improves evidence prioritization.

4.3.3 Impact of the Category-Aware FACT Inspector. We further evaluate the effectiveness of the proposed category-aware FACT inspector in Table 5. Incorporating the category-aware FACT Inspector consistently improves answer accuracy across all three

Table 5: Inspector ablation on InfoSeek and E-VQA, with the reranker fixed to Qwen3-VL-Reranker-8B. InfoSeek reports the official harmonic mean of Unseen-Q and Unseen-E; E-VQA reports BERT Matching (BEM).

Setting	LLaVA-MORE-8B		Qwen3-VL-8B		GPT-5-mini	
	InfoSeek	E-VQA	InfoSeek	E-VQA	InfoSeek	E-VQA
w/o Inspector	56.09	45.56	57.15	49.80	61.28	51.96
w/ Inspector	57.13	48.56	60.18	54.08	63.33	56.12

answerer backbones and both benchmarks. The gains reach up to 4.28 percentage points, with improvements observed not only on InfoSeek but also on E-VQA, showing the robustness of the proposed Inspector for evidence selection.

Additionally, Figure 5 visualizes the embedding space of FACT categories generated by the Inspector using E5-base embeddings [52], followed by K-means clustering ($k = 5$) and two-dimensional PCA projection. Notably, the three category-specific facts for the same query are in same semantic cluster, while representing different evidence categories, showing that it preserves the semantic consistency while providing complementary evidence across categories.⁶

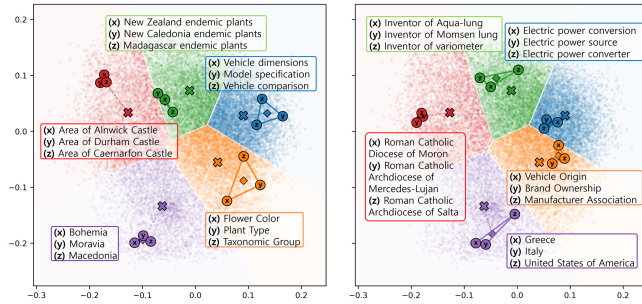


Figure 5: PCA visualization of the fact-category embedding space. 2D PCA projection and K-means clustering ($k = 5$) of E5-base embeddings with cluster regions shaded in the background. Connected circles show same-query category triplets (x, y, z) closest to (left) and farthest from (right) their respective cluster centroids in the original E5 embedding space.

5 Qualitative Analysis

To complement the quantitative evaluation, we conduct two human studies with five participants to assess: (1) whether **F²ACT-VQA** evidence improves the answerability of knowledge-intensive visual questions, and (2) whether its category-aware evidence selection aligns with human judgments.⁷

Human Evaluation of Evidence Answerability. Each participant answers **30 image-question pairs** in a multiple-choice format, with ten questions assigned to each of three conditions: **Zero-shot**,

⁶An additional ablation comparing FACTs with Wikipedia sections will be provided in a forthcoming technical supplement.

⁷Detailed study protocols and analyses will be released in a technical supplement.

Table 6: Human evaluation of evidence answerability. Acc., Grd., Attr., and Time denote accuracy, grounded accuracy, attribution, and response time. * indicates $p_{\text{Holm}} < .05$ (Acc. vs. Zero-shot; Grd./Attr. vs. RAG). Accuracy, grounded accuracy, and attribution are computed separately for each participant and then macro-averaged. Time denotes the total completion time for the 10-question block in each condition.

Condition	Acc. (%) $\uparrow$	Grd. (%) $\uparrow$	Attr. (%) $\uparrow$	Time (min) $\downarrow$
Zero-shot	30.0 $\pm$ 7.1	—	—	2.0 $\pm$ 0.1
RAG	52.0 $\pm$ 11.0	30.0 $\pm$ 10.0	56.7 $\pm$ 9.1	11.0 $\pm$ 0.6
F²ACT-VQA	78.0 $\pm$ 8.4*	66.0 $\pm$ 8.9*	84.6 $\pm$ 5.5*	2.4 $\pm$ 0.1

RAG, and **F²ACT-VQA**. For each question, the ground-truth answer is presented alongside randomly generated distractors that are semantically similar to it. Across the three conditions, participants are provided with no evidence, conventional RAG evidence, and **F²ACT-VQA** evidence, respectively, and are not allowed to access external resources. For the RAG and **F²ACT-VQA** conditions, participants additionally indicate whether their answers are based on the provided evidence or on prior knowledge. We measure accuracy, grounded accuracy, attribution, and response time, where grounded accuracy counts answers that are both correct and self-reported as relying on the provided evidence, while attribution measures evidence reliance among correct answers.

As shown in Table 6, **F²ACT-VQA** achieves the highest accuracy (78.0%), outperforming RAG (52.0%) and Zero-shot (30.0%). It also substantially improves grounded accuracy (66.0% vs. 30.0%) and attribution (84.6% vs. 56.7%) over RAG. Despite using evidence, **F²ACT-VQA** requires only 2.4 minutes, close to Zero-shot (2.0) and substantially faster than RAG (11.0), indicating that fact-centric evidence improves both answerability and evidence-use efficiency.

Human-MLLM Alignment in Evidence Selection. For each of ten instances, participants receive the top-20 retrieved FACTs and six predefined semantic categories. They select the three categories that best characterize the FACTs and one representative FACT for each category. We compare the participants’ selections with the MLLM’s original selections using category-level agreement and end-to-end representative-FACT agreement, where the latter requires an exact match in both the selected category and its representative FACT.

As shown in Figure 7, human-MLLM agreement averages 76.0% at the category level and 54.7% under the stricter end-to-end representative FACT criterion. These results suggest that the MLLM effectively captures the broad semantic organization of the retrieved evidence, while identifying the exact FACT that best represents each category remains a more fine-grained challenge.

6 Discussion

6.1 Case Study

Figure 6 illustrates how **F²ACT-VQA** addresses recurring failures in knowledge-intensive VQA.⁸ The baselines often retrieve topically relevant information tied to the wrong entity, relation, predicate,

⁸Detailed cases of baseline errors and evidence selection are provided in Figure 7.

(A) Property confusion

Q: Who is the main building contractor of this building?

Query Entity : Rundāle Palace



OMGM: Bartolomeo Rastrelli ✗
 IBA: Bartolomeo Rastrelli ✗
 EchoSight: Gaetano Chiaveri and Mikhail Zemtsov ✗

F²ACT-VQA (Ours):
 Ernst Johann von Biron ✓

Selected Evidence FACTS

- ★ [#1] Rundāle Palace | main building contractor | Ernst Johann von Biron
 [#6] Rundāle Palace | architectural style | Baroque
 [#13] Rundāle Palace | country | Latvia

(B) Entity misidentification

Q: Which road, railway or canal does this bridge carry?

Query Entity : Kessock Bridge



OMGM: A5154 ✗
 IBA: Berwickshire Railway ✗
 EchoSight: State Highway ✗

F²ACT-VQA (Ours):
 A9 road ✓

Selected Evidence FACTS

- ★ [#1] Kessock Bridge | carries | A9 road
 [#8] Kessock Bridge | location | Inverness
 [#19] Kessock Bridge | structural engineer | Hellmut Homberg

(C) Relation confusion

Q: What is the source of energy of this vehicle?

Query Entity : steam locomotive



OMGM: Steam ✗
 IBA: Steam ✗
 EchoSight: Steam ✗

F²ACT-VQA (Ours):
 coal ✓

Selected Evidence FACTS

- ★ [#1] steam locomotive | source of energy | coal
 [#3] steam locomotive | propulsion | steam
 [#13] steam locomotive | vehicle type | locomotive

(D) Temporal fact distinction

Q: In which year was this lighthouse built?

Query Entity : Umpqua River Light



OMGM: 1892 ✗
 IBA: 1894 ✗
 EchoSight: 1894 ✗

F²ACT-VQA (Ours):
 1855 ✓

Selected Evidence FACTS

- ★ [#1] first Umpqua River Light | construction year | 1855
 [#3] first Umpqua River Light | lighting year | 1857
 [#5] Umpqua River Light | heritage designation | NRHP listed place

(E) Visual entity disambiguation

Q: How tall is this building?

Query Entity : New York Marriott Marquis



OMGM: 1,776 feet ✗
 IBA: Empire State Building ✗
 EchoSight: 850 feet ✗

F²ACT-VQA (Ours):
 574 feet ✓

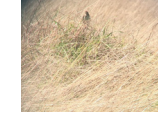
Selected Evidence FACTS

- ★ [#1] New York Marriott Marquis | height | 574 feet
 [#3] 4 World Trade Center | height | 297.7 m
 [#5] Woolworth Building | height | 351 m

(F) Fine-grained attribute grounding

Q: What is the flight call of this bird?

Query Entity : Savannah Sparrow



OMGM: zitting sound ✗
 IBA: water pipit call ✗
 EchoSight: not explicit ✗

F²ACT-VQA (Ours):
 a thin seep ✓

Selected Evidence FACTS

- ★ [#1] Savannah sparrow | flight call | a thin seep
 [#3] Savannah sparrow | flight call description | thin
 [#9] Savannah sparrow | flight feather color | brown

Figure 6: Case studies of F²ACT-VQA. Across diverse failure modes of existing knowledge-intensive VQA methods, F²ACT-VQA retrieves fine-grained, entity-grounded SPO facts that directly capture the relation required by the question, enabling accurate and interpretable answer generation. “#” indicates the rank in the RRF-fused FACT list.

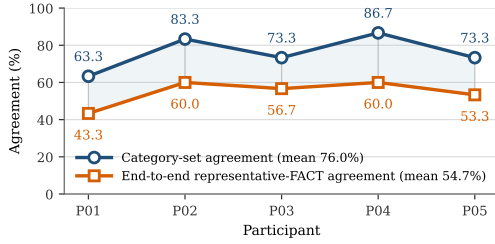


Figure 7: Human-MLLM agreement in category-aware FACT selection. Agreement is measured at the category level and with the stricter end-to-end representative FACT criterion.

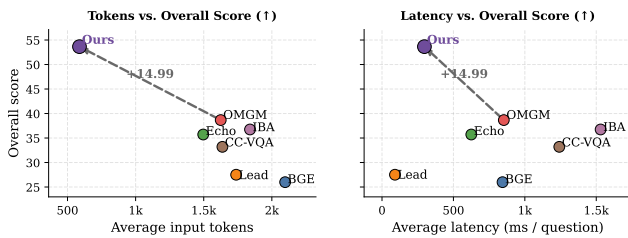


Figure 8: Averaged across datasets and backbones, F²ACT-VQA achieves the highest accuracy with the fewest input tokens while maintaining low inference latency. The left panel shows token cost and the right latency.

or time, whereas F²ACT-VQA selects compact subject-predicate-object FACTs that directly connect the visual entity to the question, especially useful for distinguishing similar entities and answering

fine-grained questions whose answers are not visually observable. The selected FACTs also make the answer basis transparent alongside plausible alternatives. Although these cases are illustrative, they qualitatively support structured, entity-grounded evidence as a means of reducing retrieval-induced confusion.

6.2 Performance-Efficiency Trade-Off

Performance-Cost Trade-Off & Inference Latency. As shown in Figure 8, F²ACT-VQA achieves the highest overall accuracy of 53.65% with the fewest answer-generation tokens. Unlike baselines that rely on longer sections or additionally processed contexts, F²ACT-VQA provides a compact set of question-relevant atomic FACTs through category-aware inspection. This yields competitive latency while indicating that the gain comes from more precise rather than more extensive evidence.

Inference Latency. F²ACT-VQA also maintains low inference latency at approximately 297 ms per query, reducing latency by about 68% on average compared with retrieval- and reasoning-intensive baselines such as EchoSight, OMGM, IBA, CC-VQA, and Naive-RAG-BGE. Although Naive-RAG-Lead is faster, its accuracy is considerably lower. Overall, F²ACT-VQA provides a favorable balance between answer quality and computational cost.

7 Conclusion

We present F²ACT-VQA, a fusion-driven, fact-centric retrieval-augmented framework for knowledge-intensive VQA that represents external knowledge as Subject-Predicate-Object (SPO) triplets. F²ACT-VQA performs global fact retrieval via weighted fusion of visual features and structured SPO components, and refines the retrieved candidates with multimodal MLLM reranking. Experiments on knowledge-intensive VQA benchmarks show that this design

improves both retrieval quality and downstream answer accuracy, highlighting structured fact-level retrieval as a practical mechanism for grounding MLLMs in external knowledge.

Use of AI Assistants

We used general-purpose AI assistants only for feedback on writing quality and clarity, and not for experiment design, data analysis, or result generation. All scientific content was written and revised by the authors.

Ethical Considerations

F²ACT-VQA relies on external knowledge sources and pretrained multimodal models, and may therefore inherit factual errors, outdated information, or biases from them. Although the fact-centric design makes supporting evidence more explicit, errors in FACT generation, retrieval, or reranking may still propagate to the final answer. In high-stakes settings, retrieved evidence should therefore be independently verified and model outputs used with appropriate human oversight. Representing document-level knowledge as atomic SPO facts may also remove contextual details or nuance from the original source. Moreover, category-aware evidence selection necessarily retains only a subset of retrieved facts, which may omit relevant contextual or complementary information when the underlying knowledge is ambiguous or incomplete. Such errors could lead to seemingly well-grounded but misleading answers. **F²ACT-VQA** is therefore intended as a research framework rather than a decision-making system for sensitive domains.

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